

Flask - MySQL

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 - ✓ Conectare Python la MySQL
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 - ✓ Creare tabelă
 - ✓ Inserare date de test
 - ✓ Citire date din tabelă
 - ✓ Verificări activități suspecte MySQL
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 - ✓ Decriptare date în browser
-

The MySQL logo is displayed in a large, orange, sans-serif font. It is centered within a light orange rectangular area that has rounded corners and is enclosed by a dark blue dashed border.

MySQL

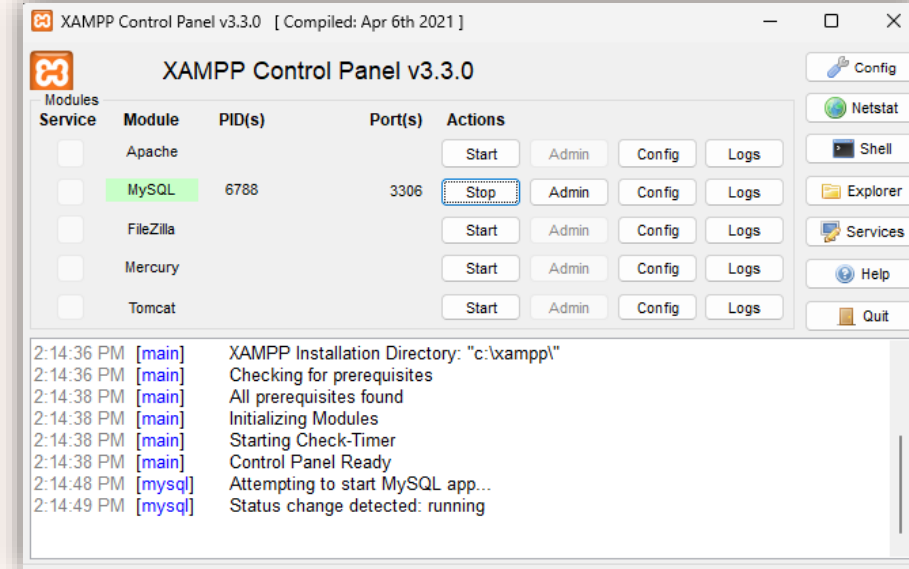
Structured Query Language

SQL este limbajul prin care dai comenzi unei baze de date.

Dacă nu scriem portul, MySQL Connector folosește implicit 3306

```
25
26 conn = mysql.connector.connect(
27     host="localhost",
28     user="root",
29     password="",
30     database="nbd_bank"
31 )
32
```

Connector = biblioteca/driverul care face legătura între Python și MySQL.



pip install mysql-connector-python

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> pip install mysql-connector-python
Collecting mysql-connector-python
  Downloading mysql_connector_python-9.7.0-cp313-cp313-win_amd64.whl.metadata (11 kB)
  Downloading mysql_connector_python-9.7.0-cp313-cp313-win_amd64.whl (17.7 MB)
    17.7/17.7 MB 9.3 MB/s 0:00:01
Installing collected packages: mysql-connector-python
Successfully installed mysql-connector-python-9.7.0

[notice] A new release of pip is available: 25.2 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
```

```
25
26 conn = mysql.connector.connect(
27     host="localhost",
28     port=3306,
29     user="root",
30     password="",
31     database="nbd_bank"
32 )
33
```

C: > Users > Paul > Desktop > f.py > ...

```
1  import mysql.connector
2
3  # 1. Conectare la MySQL
4  conn = mysql.connector.connect(
5      host="localhost",
6      user="root",
7      password=""
8  )
9
10 cursor = conn.cursor()
11
12 # 2. Creare bază de date
13 cursor.execute("CREATE DATABASE IF NOT EXISTS nbd_bank")
14
15 # 3. Selectare bază de date
16 cursor.execute("USE nbd_bank")
17
18 # 4. Creare tabelă
19 cursor.execute("""
20 CREATE TABLE IF NOT EXISTS transactions (
21     id INT AUTO_INCREMENT PRIMARY KEY,
22     client VARCHAR(50),
23     amount FLOAT,
24     country VARCHAR(10),
25     hour INT
26 )
27 """)
```

“cursor” - numele vine de la ideea de indicator care se mută prin rezultate.

INT	-- număr întreg
FLOAT	-- număr cu zecimale
VARCHAR	-- text scurt
TEXT	-- text lung
DATE	-- dată
DATETIME	-- dată + oră
BOOLEAN	-- true/false

execută această comandă SQL în MySQL.

“transactions” este numele tabelului.

Creează coloana id.

Creează coloana client. VARCHAR(50) înseamnă text de maximum 50 de caractere.

număr real, cu zecimale.

număr întreg.

```
28
29 # 5. Date de test
30 data = [
31     ("C1", 120, "RO", 14),
32     ("C2", 15000, "RU", 2),
33     ("C3", 80, "RO", 11),
34     ("C4", 9700, "NG", 3),
35     ("C5", 300, "FR", 21)
36 ]
37
38 # 6. Inserare date
39 cursor.executemany("""
40 INSERT INTO transactions (client, amount, country, hour)
41 VALUES (%s, %s, %s, %s)
42 """, data)
43
44 conn.commit()
45
```

nbd_bank = baza de date
transactions = tabela
client = coloană
amount = coloană
country = coloană
hour = coloană

executemany() înseamnă: execută INSERT-ul de mai multe ori, câte o dată pentru fiecare rând din var. data.

%s sunt placeholder, adica locuri rezervate pentru valorile din fiecare rând din data.

Daca nu vrem tot (*), putem scrie doar numele coloanelor dorite, ex:
SELECT client, amount FROM transactions

in SQL, * înseamnă tot.

fetchall() = ia toate rândurile returnate de MySQL

RU = Rusia
NG = Nigeria
KP = Coreea de Nord

row = o înregistrare / un rând din tabelă.

```
45 # 7. Citire date din tabelă
46 cursor.execute("SELECT * FROM transactions")
47
48
49 rows = cursor.fetchall()
50
51 print("Tranzacții salvate în MySQL:")
52
53 for row in rows:
54     print(row)
55
56 # 8. Închidere conexiune
57 cursor.close()
58 conn.close()
```

cursor = obiect local în Python

Tranzacții salvate în MySQL:

```
(1, 'C1', 120.0, 'RO', 14)
(2, 'C2', 15000.0, 'RU', 2)
(3, 'C3', 80.0, 'RO', 11)
(4, 'C4', 9700.0, 'NG', 3)
(5, 'C5', 300.0, 'FR', 21)
```

PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>



Verificari activitati
suspecte MySQL


```

1 import mysql.connector
2
3 conn = mysql.connector.connect(
4     host="localhost",
5     user="root",
6     password="",
7     database="nbd_bank"
8 )
9
10 cursor = conn.cursor()
11
12 query = """
13 SELECT id, client, amount, country, hour
14 FROM transactions
15 WHERE amount > 5000
16     OR hour < 5
17     OR country IN ('RU', 'NG', 'KP')
18 """
19
20 cursor.execute(query)
21
22 rows = cursor.fetchall()
23
24 print("Tranzacții suspecte:")
25
26 for row in rows:
27     print(row)
28
29 cursor.close()
30 conn.close()

```

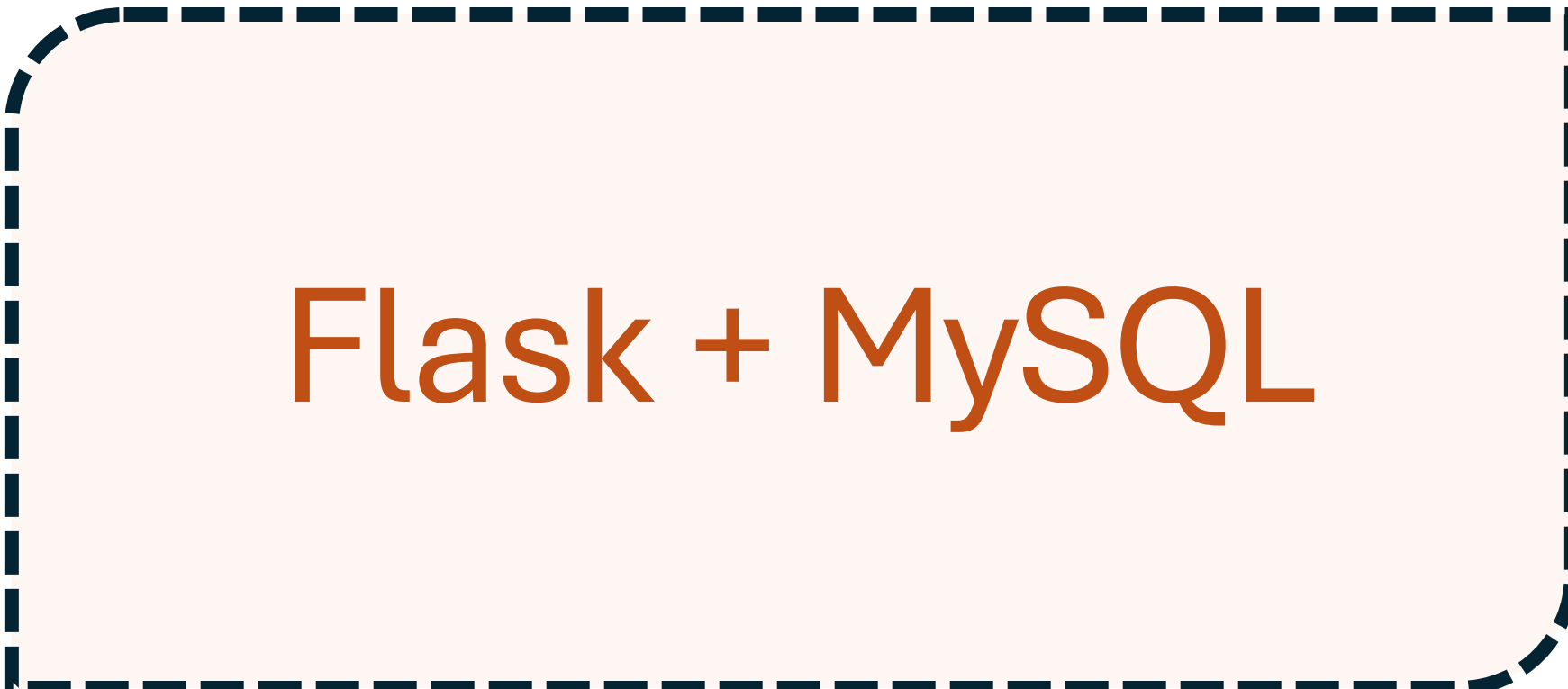
Comanda selectează coloanele id, client, amount, country, hour din tabela transactions. Apoi filtrează doar tranzacțiile unde amount > 5000, sau hour < 5, sau country este una dintre valorile RU, NG, KP. Cu alte cuvinte, scoate tranzacțiile considerate suspecte după aceste reguli.

```

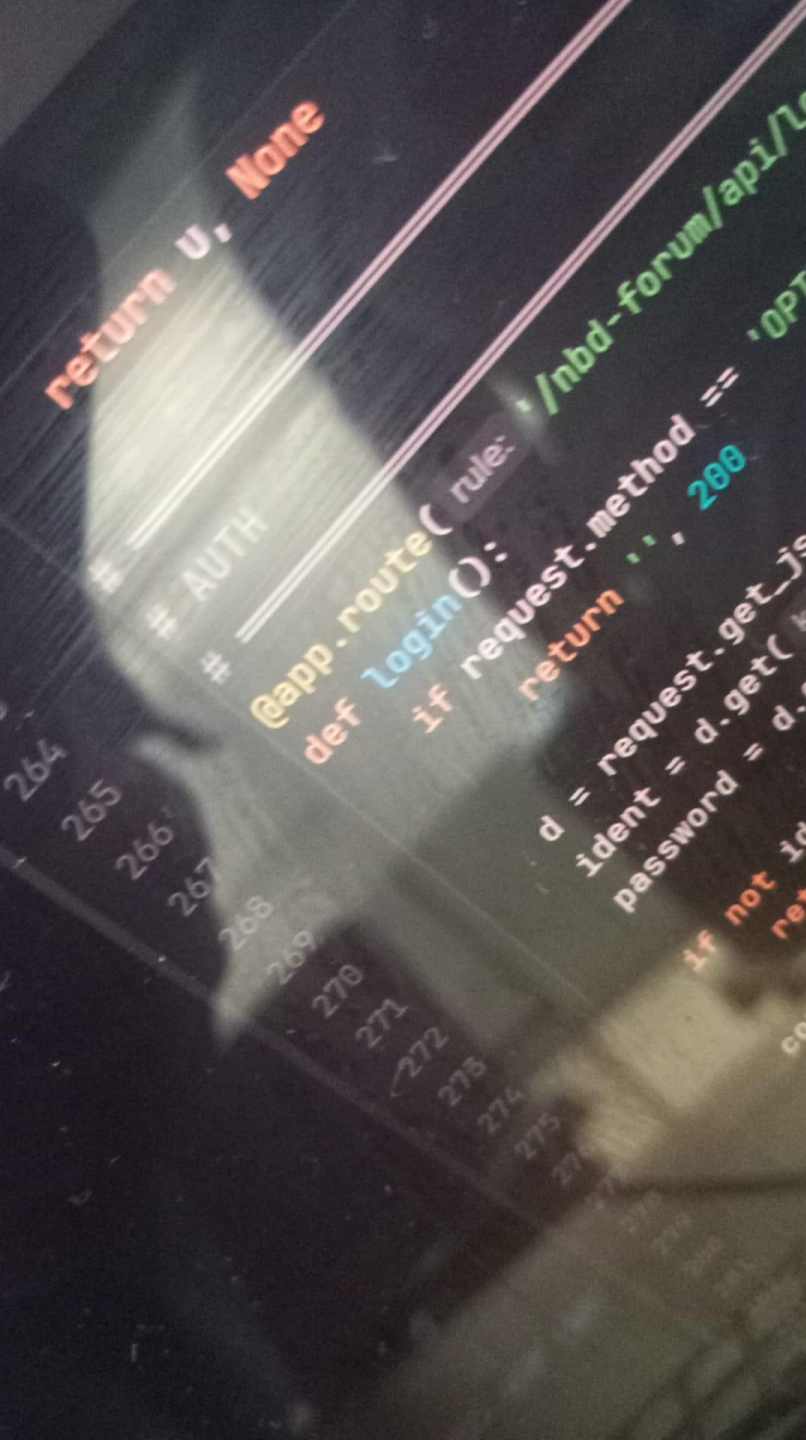
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
Tranzacții suspecte:
(2, 'C2', 15000.0, 'RU', 2)
(4, 'C4', 9700.0, 'NG', 3)
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>

```

RU = Rusia
 NG = Nigeria
 KP = Coreea de Nord



Flask + MySQL



```
pip install flask
```

```
python -m flask --version
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Requirement already satisfied: colorama in c:\users\paul\appdata\
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.4.1-py3-none-any.whl (116 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading markupsafe-3.0.3-cp313-cp313-win_amd64.whl (15 kB)
Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
Installing collected packages: markupsafe, itsdangerous, click, b
Successfully installed blinker-1.9.0 click-8.4.1 flask-3.1.3 itsd

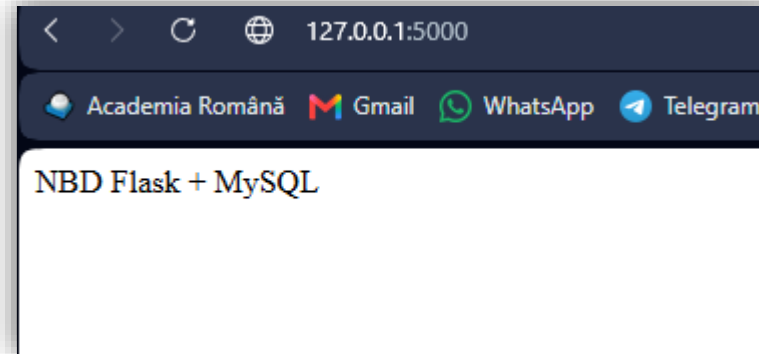
[notice] A new release of pip is available: 25.2 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
```

Flask instalare

```
[notice] A new release of pip is available: 25.2 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> python -m flask --version
Python 3.13.7
Flask 3.1.3
Werkzeug 3.1.8
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
```

Pasul 0 - scheletul minim Flask

```
1  from flask import Flask
2  import mysql.connector
3
4  app = Flask(__name__)
5
6  def db():
7      return mysql.connector.connect(
8          host="localhost",
9          user="root",
10         password="",
11         database="nbd_bank"
12     )
13
14  @app.route("/")
15  def home():
16      return "NBD Flask + MySQL"
17
18  app.run(debug=True)
```



```
PS C:\Users\Paul\AppData\Local\Program
* Serving Flask app 'f'
* Debug mode: on
WARNING: This is a development server.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
```

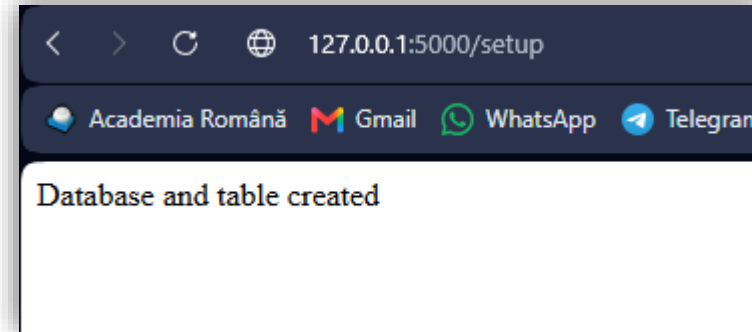


Pasul 1 - endpoint pentru creare bază și tabelă

Adaugati acest cod înainte de `app.run(debug=True)`:

```
19 @app.route("/setup")
20 def setup():
21     conn = mysql.connector.connect(
22         host="localhost",
23         user="root",
24         password=""
25     )
26
27     cursor = conn.cursor()
28
29     cursor.execute("CREATE DATABASE IF NOT EXISTS nbd_bank")
30     cursor.execute("USE nbd_bank")
31
32     cursor.execute("""
33     CREATE TABLE IF NOT EXISTS transactions (
34         id INT AUTO_INCREMENT PRIMARY KEY,
35         client VARCHAR(50),
36         amount FLOAT
37     )
38     """)
39
40     conn.commit()
41     conn.close()
42
43     return "Database and table created"
```

`http://127.0.0.1:5000/setup`



```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
* Serving Flask app 'f'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 15:55:29] "GET /setup HTTP/1.1" 200 -
█
```

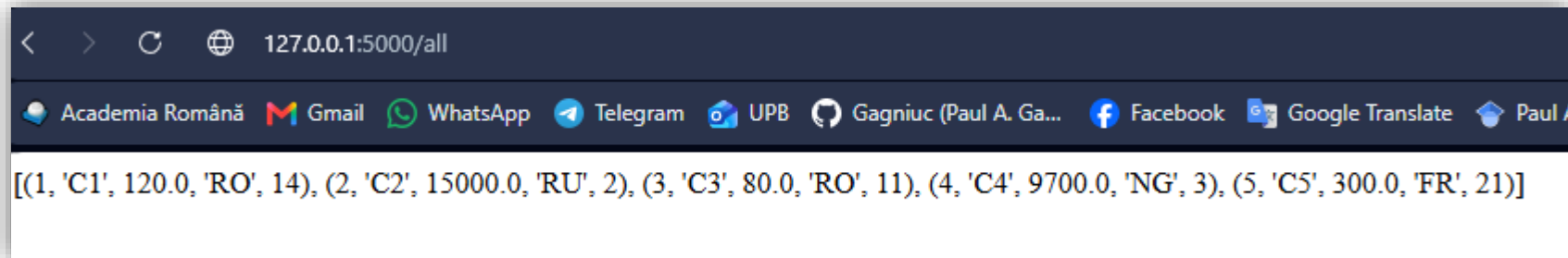
Pasul 2 - endpoint pentru afișarea datelor

Adaugati acest cod înainte de `app.run(debug=True)`:

```
46
47 @app.route("/all")
48 def all_data():
49     conn = db()
50     cursor = conn.cursor()
51
52     cursor.execute("SELECT * FROM transactions")
53     rows = cursor.fetchall()
54
55     conn.close()
56
57     return str(rows)
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:
* Serving Flask app 'f'
* Debug mode: on
WARNING: This is a development server. Do not use it in a produ
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 16:00:50] "GET /all HTTP/1.1" 200 -
```

<http://127.0.0.1:5000/all>



La început trebuie să apară: `[]`, dacă tabela este goală. Totuși, în cazul nostru, tabela există deja din exercițiul precedent, deci este posibil să apară înregistrările introduse anterior.

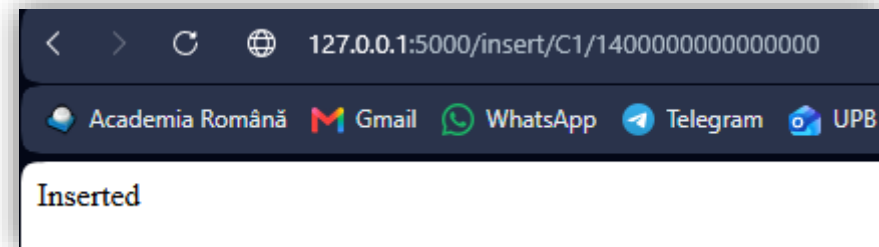
Adaugati acest cod înainte de `app.run(debug=True)`:

```
59
60 @app.route("/insert/<client>/<amount>")
61 def insert(client, amount):
62     conn = db()
63     cursor = conn.cursor()
64
65     cursor.execute(
66         "INSERT INTO transactions(client, amount) VALUES(%s, %s)",
67         (client, amount)
68     )
69
70     conn.commit()
71     conn.close()
72
73     return "Inserted"
```

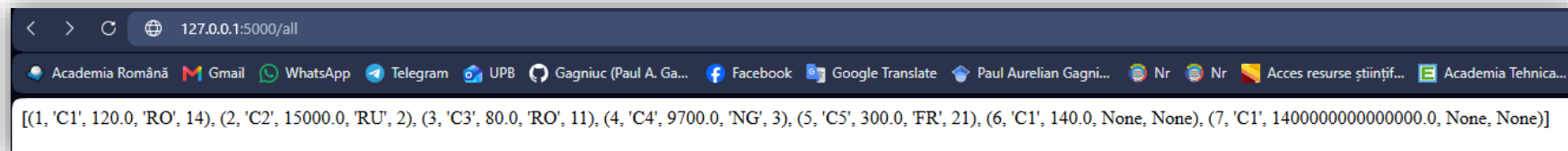
```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
* Serving Flask app 'f'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production grade web server.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 16:52:36] "GET /insert/C1/140 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 16:53:03] "GET /all HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 16:53:04] "GET /all HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 16:53:32] "GET /insert/C1/1400000000000000 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 16:53:37] "GET /all HTTP/1.1" 200 -
```

Pasul 3 - endpoint pentru inserare

<http://127.0.0.1:5000/insert/C1/140000000000>



Deci “country” și “hour” rămân goale.

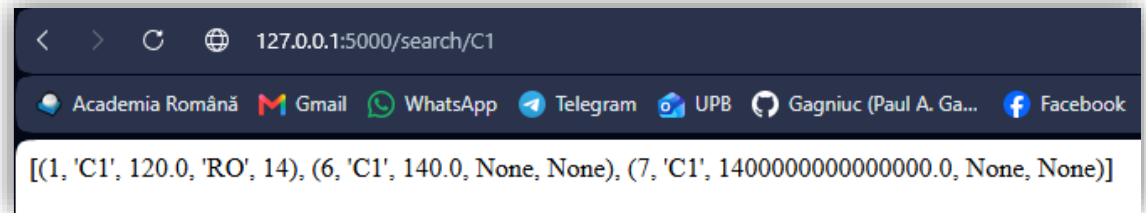


Adaugati acest cod înainte de `app.run(debug=True)`:

```
75
76 @app.route("/search/<client>")
77 def search(client):
78     conn = db()
79     cursor = conn.cursor()
80
81     cursor.execute(
82         "SELECT * FROM transactions WHERE client=%s",
83         (client,)
84     )
85
86     rows = cursor.fetchall()
87     conn.close()
88
89     return str(rows)
90
```

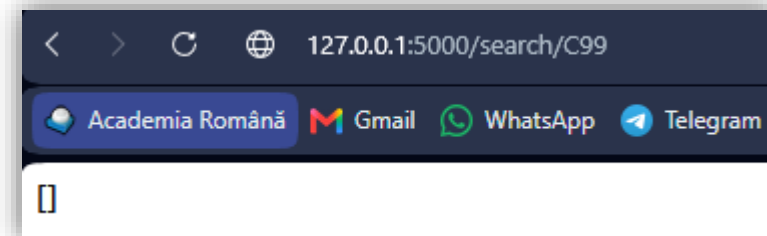
Pasul 4 - endpoint pentru căutare client

`http://127.0.0.1:5000/search/C1`



Dacă nu există:

`http://127.0.0.1:5000/search/C99`



```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
* Serving Flask app 'f'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 17:00:19] "GET /search/C1 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 17:01:34] "GET /search/C99 HTTP/1.1" 200 -

```

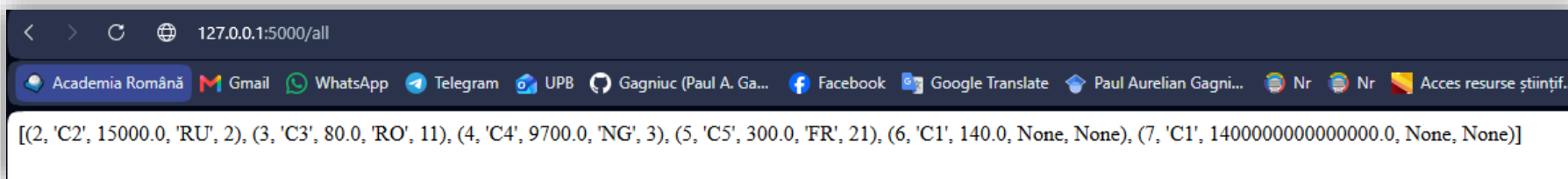

Adaugați acest cod înainte de `app.run(debug=True)`:

```
91
92 @app.route("/delete/<id>")
93 def delete(id):
94     conn = db()
95     cursor = conn.cursor()
96
97     cursor.execute(
98         "DELETE FROM transactions WHERE id=%s",
99         (id,)
100    )
101
102    conn.commit()
103    conn.close()
104
105    return "Deleted"
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\
* Serving Flask app 'f'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 18:14:01] "GET /delete/1 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 18:16:03] "GET /all HTTP/1.1" 200 -
[]
```

Dupa:

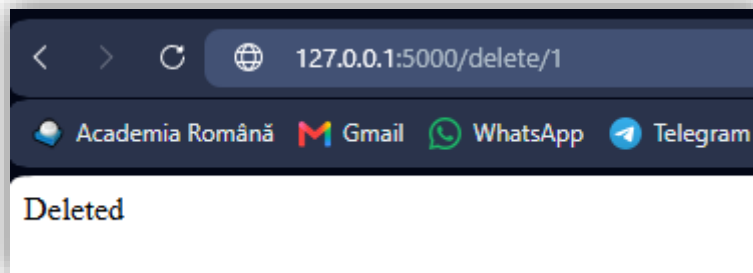
`http://127.0.0.1:5000/all`



```
[
  (2, 'C2', 15000.0, 'RU', 2),
  (3, 'C3', 80.0, 'RO', 11),
  (4, 'C4', 9700.0, 'NG', 3),
  (5, 'C5', 300.0, 'FR', 21),
  (6, 'C1', 140.0, None, None),
  (7, 'C1', 1400000000000000.0, None, None)
]
```

Pasul 5 - endpoint pentru ștergere

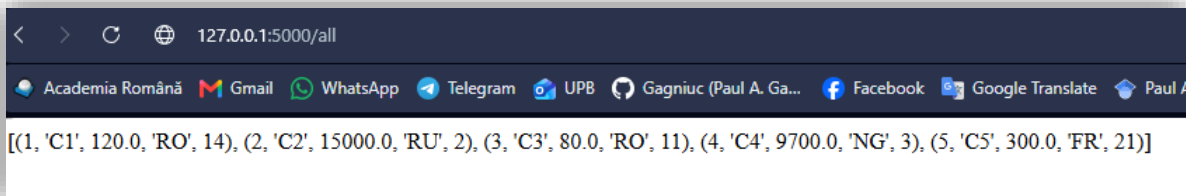
`http://127.0.0.1:5000/delete/1`



```
Deleted
```

Înainte:

`http://127.0.0.1:5000/all`



```
[
  (1, 'C1', 120.0, 'RO', 14),
  (2, 'C2', 15000.0, 'RU', 2),
  (3, 'C3', 80.0, 'RO', 11),
  (4, 'C4', 9700.0, 'NG', 3),
  (5, 'C5', 300.0, 'FR', 21)
]
```

Tranzacția cu id = 1 nu mai apare.

Adaugati acest cod înainte de `app.run(debug=True)`:

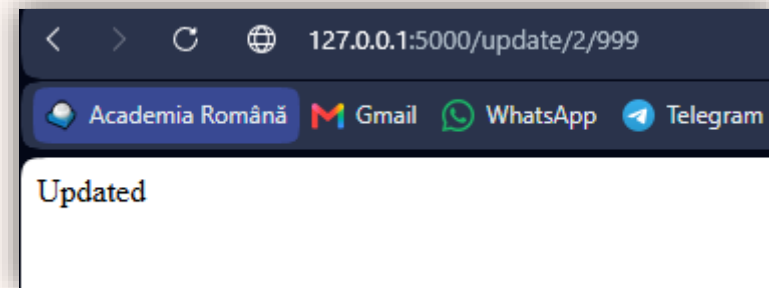
```
107
108 @app.route("/update/<id>/<amount>")
109 def update(id, amount):
110     conn = db()
111     cursor = conn.cursor()
112
113     cursor.execute(
114         "UPDATE transactions SET amount=%s WHERE id=%s",
115         (amount, id)
116     )
117
118     conn.commit()
119     conn.close()
120
121     return "Updated"
122
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
* Serving Flask app 'f'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 18:24:07] "GET /update/2/999 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 18:28:07] "GET /all HTTP/1.1" 200 -

```

Pasul 6 - endpoint pentru modificare

`http://127.0.0.1:5000/update/2/999`



`http://127.0.0.1:5000/all`

Trebuie să vezi că suma pentru id = 2 a devenit 999.



Pasul 7 - ștergere rapidă a tuturor rândurilor

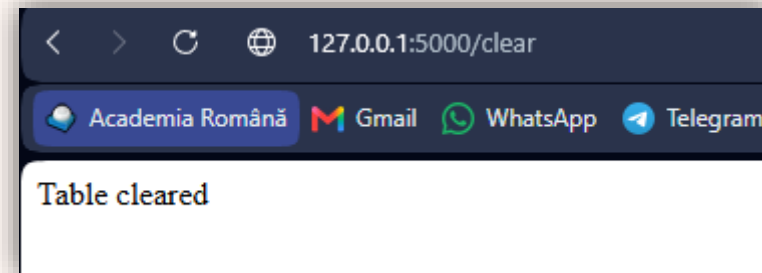
Adaugati acest cod înainte de `app.run(debug=True)`:

```
124
125 @app.route("/clear")
126 def clear():
127     conn = db()
128     cursor = conn.cursor()
129
130     cursor.execute("TRUNCATE TABLE transactions")
131
132     conn.commit()
133     conn.close()
134
135     return "Table cleared"
136
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\
* Serving Flask app 'Flask_MySQL'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 19:06:03] "GET /clear HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 19:07:17] "GET /all HTTP/1.1" 200 -
█
```

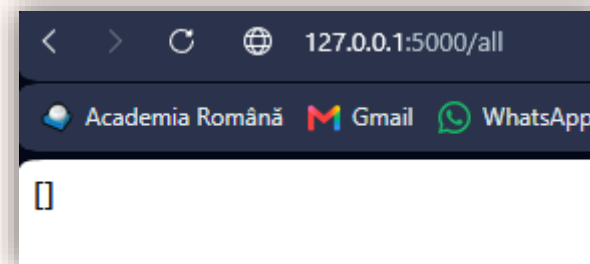
Înainte:

`http://127.0.0.1:5000/clear`



După:

`http://127.0.0.1:5000/clear`



TRUNCATE TABLE transactions șterge **toate rândurile** din tabela transactions.

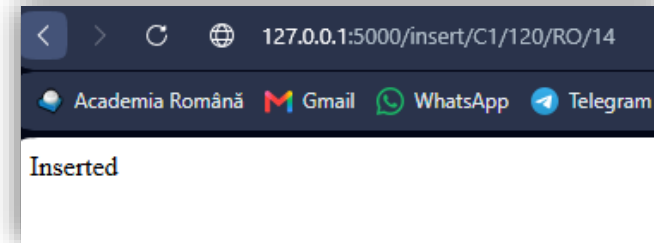
Pasul 8 - endpoint insert corect/complet pentru tabela actuală

Adaugati acest cod înainte de `app.run(debug=True)`:

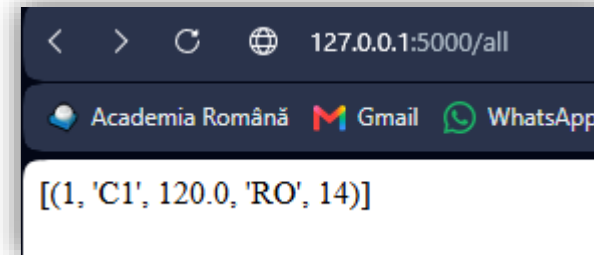
```
138 @app.route("/insert/<client>/<amount>/<country>/<hour>")
139 def insert(client, amount, country, hour):
140     conn = db()
141     cursor = conn.cursor()
142
143     cursor.execute(
144         "INSERT INTO transactions(client, amount, country, hour) VALUES(%s, %s, %s, %s)",
145         (client, amount, country, hour)
146     )
147
148     conn.commit()
149     conn.close()
150
151     return "Inserted"
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
* Serving Flask app 'Flask_MySQL'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 19:18:14] "GET /insert/C1/120/RO/14 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 19:19:21] "GET /all HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 19:20:01] "GET /insert/C1/120/RO/14 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 19:20:04] "GET /all HTTP/1.1" 200 -
```

<http://127.0.0.1:5000/insert/C1/120/RO/14>

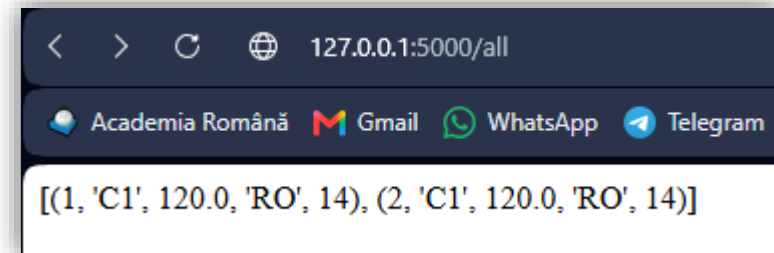


<http://127.0.0.1:5000/all>



2 x <http://127.0.0.1:5000/insert/C1/120/RO/14>

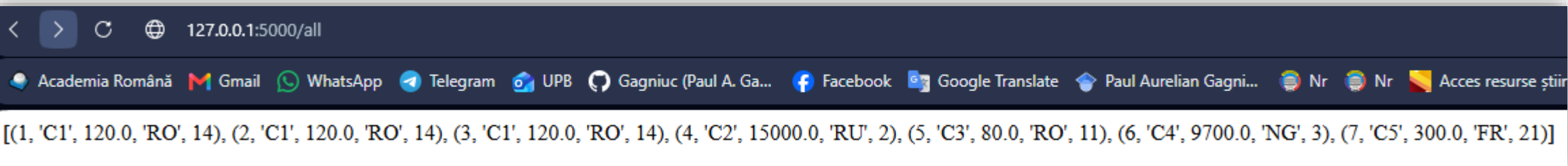
<http://127.0.0.1:5000/all>



Adaugati acest cod înainte de `app.run(debug=True)`:

```
154
155 @app.route("/seed")
156 def seed():
157     conn = db()
158     cursor = conn.cursor()
159
160     data = [
161         ("C1", 120, "RO", 14),
162         ("C2", 15000, "RU", 2),
163         ("C3", 80, "RO", 11),
164         ("C4", 9700, "NG", 3),
165         ("C5", 300, "FR", 21)
166     ]
167
168     cursor.executemany(
169         "INSERT INTO transactions(client, amount, country, hour) VALUES(%s,%s,%s,%s)",
170         data
171     )
172
173     conn.commit()
174     conn.close()
175
176     return "Test data inserted"
177
```

<http://127.0.0.1:5000/all>

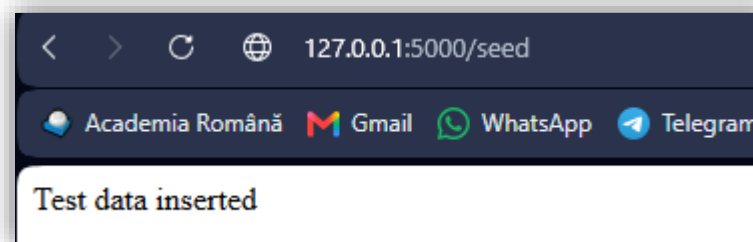


[(1, 'C1', 120.0, 'RO', 14), (2, 'C1', 120.0, 'RO', 14), (3, 'C1', 120.0, 'RO', 14), (4, 'C2', 15000.0, 'RU', 2), (5, 'C3', 80.0, 'RO', 11), (6, 'C4', 9700.0, 'NG', 3), (7, 'C5', 300.0, 'FR', 21)]

Pasul 9 - încarcă date de test automat

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\
* Serving Flask app 'Flask_MySQL'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 21:26:33] "GET /seed HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 21:27:19] "GET /all HTTP/1.1" 200 -
```

<http://127.0.0.1:5000/seed>



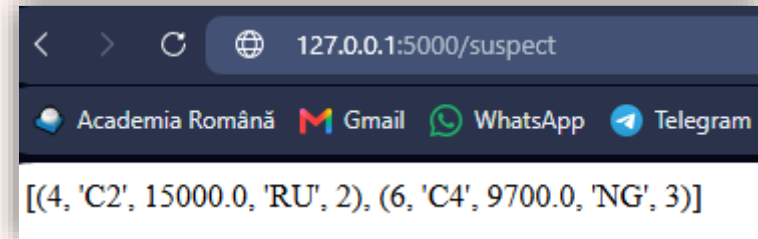
Test data inserted

Pasul 10 - tranzacții suspecte

```
178
179
180 @app.route("/suspect")
181 def suspect():
182     conn = db()
183     cursor = conn.cursor()
184
185     cursor.execute("""
186         SELECT * FROM transactions
187         WHERE amount > 5000 OR hour < 5 OR country IN ('RU', 'NG', 'KP')
188     """)
189
190     rows = cursor.fetchall()
191     conn.close()
192
193     return str(rows)
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\
* Serving Flask app 'Flask_MySQL'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 21:33:07] "GET /suspect HTTP/1.1" 200 -
█
```

http://127.0.0.1:5000/suspect



Pasul 11 - raport rapid din baza de date

```
195
196
197 @app.route("/stats")
198 def stats():
199     conn = db()
200     cursor = conn.cursor()
201
202     cursor.execute("""
203         SELECT
204             COUNT(*) AS total_tranzactii,
205             SUM(amount) AS suma_totala,
206             AVG(amount) AS media
207         FROM transactions
208     """)
209
210     row = cursor.fetchone()
211     conn.close()
212
213     return str(row)
214
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
* Serving Flask app 'Flask_MySQL'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 21:45:00] "GET /stats HTTP/1.1" 200 -
```

http://127.0.0.1:5000/stats

(7, 25440.0, 3634.285714285714)

COUNT(*) = numără rândurile din tabelă.
SUM(amount) = adună toate sumele.
AVG(amount) = calculează media sumelor.
AS = dă un nume rezultatului.

http://127.0.0.1:5000/all

[(1, 'C1', 120.0, 'RO', 14), (2, 'C1', 120.0, 'RO', 14), (3, 'C1', 120.0, 'RO', 14), (4, 'C2', 15000.0, 'RU', 2), (5, 'C3', 80.0, 'RO', 11), (6, 'C4', 9700.0, 'NG', 3), (7, 'C5', 300.0, 'FR', 21)]


```

214
215
216 @app.route("/client_total/<client>")
217 def client_total(client):
218     conn = db()
219     cursor = conn.cursor()
220
221     cursor.execute(
222         "SELECT client, SUM(amount) FROM transactions WHERE client=%s GROUP BY client",
223         (client,)
224     )
225
226     row = cursor.fetchone()
227     conn.close()
228
229     return str(row)
230

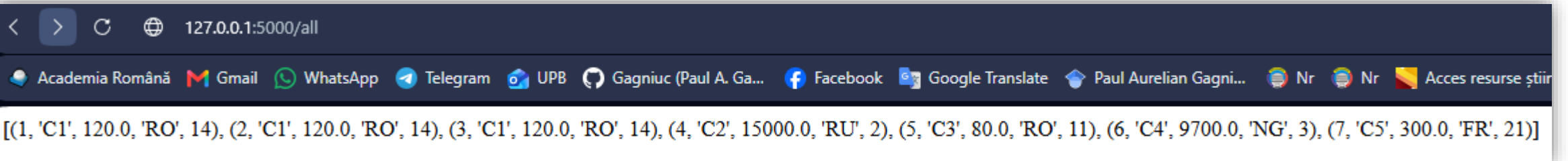
```

```

PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
* Serving Flask app 'Flask_MySQL'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 21:50:53] "GET /client_total/C1 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 21:58:02] "GET /client_total/C1 HTTP/1.1" 200 -

```

http://127.0.0.1:5000/all



127.0.0.1:5000/all

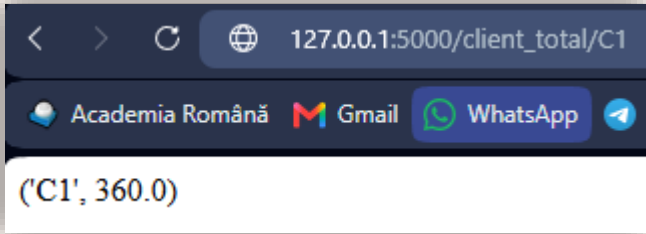
Academia Română Gmail WhatsApp Telegram UPB Gagniuc (Paul A. Ga... Facebook Google Translate Paul Aurelian Gagni... Nr Nr Acces resurse știin

[(1, 'C1', 120.0, 'RO', 14), (2, 'C1', 120.0, 'RO', 14), (3, 'C1', 120.0, 'RO', 14), (4, 'C2', 15000.0, 'RU', 2), (5, 'C3', 80.0, 'RO', 11), (6, 'C4', 9700.0, 'NG', 3), (7, 'C5', 300.0, 'FR', 21)]

Pasul 12 - total pe client

Endpointul /client_total/C1 nu afișează o singură tranzacție, ci totalul tuturor tranzacțiilor clientului C1.

http://127.0.0.1:5000/client_total/C1



(C1', 360.0)

(1, 'C1', 120.0, 'RO', 14)
 (2, 'C1', 120.0, 'RO', 14)
 (3, 'C1', 120.0, 'RO', 14)

360



Flask+MySQL+
Criptare

pip install cryptography

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> pip install cryptography
Collecting cryptography
  Downloading cryptography-49.0.0-cp311-abi3-win_amd64.whl.metadata (4.3 kB)
Requirement already satisfied: cffi>=2.0.0 in c:\users\paul\appdata\local\programs\python\python311\lib\site-packages (from cryptography)
Requirement already satisfied: pycparser in c:\users\paul\appdata\local\programs\python\python311\lib\site-packages (from cffi>=2.0.0->cryptography)
Downloading cryptography-49.0.0-cp311-abi3-win_amd64.whl (3.8 MB)
----- 3.8/3.8 MB 8.3 MB/s 0:00:00
Installing collected packages: cryptography
Successfully installed cryptography-49.0.0

[notice] A new release of pip is available: 25.2 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> |
```

C: > Users > Paul > Desktop > Flask_MySQLcript.py > ...

```
1  from cryptography.fernet import Fernet
2  import base64
3  import hashlib
4
5  def make_key(password):
6      return base64.urlsafe_b64encode(
7          hashlib.sha256(password.encode()).digest()
8      )
9
10 def encrypt(text, password):
11     f = Fernet(make_key(password))
12     return f.encrypt(str(text).encode()).decode()
13
14 def decrypt(text, password):
15     f = Fernet(make_key(password))
16     return f.decrypt(text.encode()).decode()
17
18 secret = encrypt("15000", "parola123")
19 print(secret)
20
21 normal = decrypt(secret, "parola123")
22 print(normal)
```



```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Python\Python38-32\python.exe C:\Users\Paul\Desktop\Flask_MySQLcript.py
gAAAAABqTrWwP7WBoGxWzYRK4b3om4bsChcUS2UgaO3Fe_YWeFvz_wSa0YhbBS_KBIYuwEQYaGHUwc_Px5HApppI1PwzQ9uuqA==
15000
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> █
```

Inserarea criptată

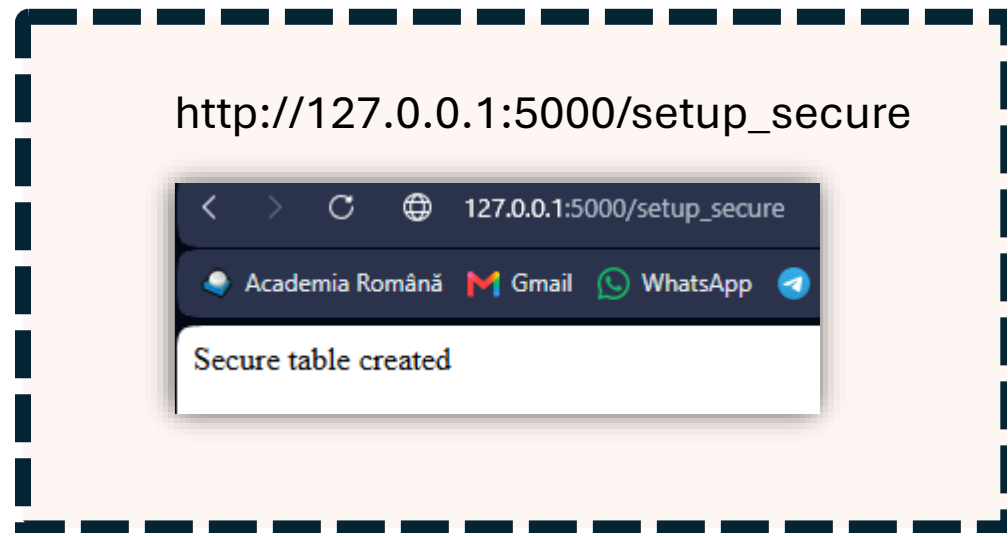
```
1 from flask import Flask
2 import mysql.connector
3
4 from cryptography.fernet import Fernet
5 import base64
6 import hashlib
7
8 def make_key(password):
9     return base64.urlsafe_b64encode(
10         hashlib.sha256(password.encode()).digest()
11     )
12
13 def encrypt(text, password):
14     f = Fernet(make_key(password))
15     return f.encrypt(str(text).encode()).decode()
16
17 def decrypt(text, password):
18     f = Fernet(make_key(password))
19     return f.decrypt(text.encode()).decode()
20
21
22 app = Flask(__name__)
23
24 def db():
25     return mysql.connector.connect(
26         host="localhost",
27         user="root",
28         password="",
29         database="nbd_bank"
30     )
31
32 @app.route("/")
33 def home():
34     return "NBD Flask + MySQL"
```



```

250 @app.route("/setup_secure")
251 def setup_secure():
252     conn = db()
253     cursor = conn.cursor()
254
255     cursor.execute("DROP TABLE IF EXISTS secure_transactions")
256
257     cursor.execute("""
258 CREATE TABLE secure_transactions (
259     id INT AUTO_INCREMENT PRIMARY KEY,
260     client VARCHAR(50),
261     amount TEXT,
262     country TEXT,
263     hour TEXT
264 )
265 """)
266
267     conn.commit()
268     conn.close()
269
270     return "Secure table created"
271

```



Pentru date criptate, tabela trebuie să fie așa:

```

id INT AUTO_INCREMENT PRIMARY KEY
client VARCHAR(50)
amount TEXT
country TEXT
hour TEXT

```

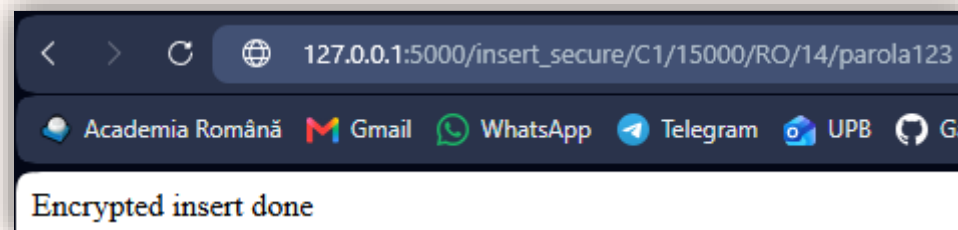
Adică toate coloanele criptate trebuie să fie TEXT.

```

275 @app.route("/insert_secure/<client>/<amount>/<country>/<hour>/<password>")
276 def insert_secure(client, amount, country, hour, password):
277     conn = db()
278     cursor = conn.cursor()
279
280
281     encrypted_amount = encrypt(amount, password)
282     encrypted_country = encrypt(country, password)
283     encrypted_hour = encrypt(hour, password)
284
285     cursor.execute(
286         "INSERT INTO secure_transactions(client, amount, country, hour) VALUES(%s, %s, %s, %s)",
287         (client, encrypted_amount, encrypted_country, encrypted_hour)
288     )
289
290     conn.commit()
291     conn.close()
292
293     return "Encrypted insert done"

```

http://127.0.0.1:5000/insert_secure/C1/15000/RO/14/parola123



```

295
296
297 @app.route("/secure_raw")
298 def secure_raw():
299     conn = db()
300     cursor = conn.cursor()
301
302     cursor.execute("SELECT * FROM secure_transactions")
303     rows = cursor.fetchall()
304
305     conn.close()
306
307     return str(rows)
308

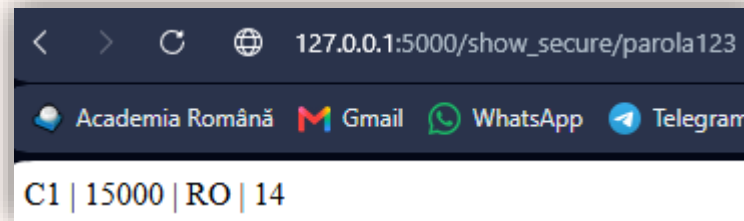
```



```
311 @app.route("/show_secure/<password>")
312 def show_secure(password):
313     conn = db()
314     cursor = conn.cursor()
315
316     cursor.execute("SELECT client, amount, country, hour FROM secure_transactions")
317     rows = cursor.fetchall()
318
319     result = ""
320
321     for client, amount, country, hour in rows:
322         amount_dec = decrypt(amount, password)
323         country_dec = decrypt(country, password)
324         hour_dec = decrypt(hour, password)
325
326         result += f"{client} | {amount_dec} | {country_dec} | {hour_dec}<br>"
327
328     conn.close()
329
330
```



http://127.0.0.1:5000/show_secure/parola123



Cautari approximative.

Ex: LIKE '%ion%' = găsește orice client care conține „ion”

```
335
336 @app.route("/search_like/<name>")
337 ✓ def search_like(name):
338     conn = db()
339     cursor = conn.cursor()
340
341 ✓     cursor.execute(
342         "SELECT * FROM transactions WHERE client LIKE %s",
343         ("% " + name + "%",)
344     )
345
346     rows = cursor.fetchall()
347     conn.close()
348
349     return str(rows)
350
```



http://127.0.0.1:5000/search_like/C

Ză end!

